

EXCEL ANALYSIS REPORT

E-Commerce Sales Dataset — Day 2

7-Day Data Analyst Portfolio Roadmap — Full Step-by-Step Documentation

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1. Introduction

This report documents the data analysis process applied to the e-commerce sales dataset that was cleaned on Day 1 (cleaned_sales_data.xlsx, 3,933 rows). The analysis was carried out entirely in Microsoft Excel using Pivot Tables, formulas (SUMIFS, COUNTIFS, IF, INDEX/MATCH), and PivotCharts, following the Day 2 checklist in the 7-day Data Analyst portfolio roadmap.

Unlike the Day 1 report, this one is written as a step-by-step tutorial — each section explains HOW to do something in Excel (which menu to click, which field to drag, which formula to type), not just the end result. The goal is so the process can be repeated or taught to someone else without having to recall everything from scratch.

Working file: Sales_Analysis.xlsx — created via Save As from cleaned_sales_data.xlsx, with additional sheets: pivot_tabels, Summary, Charts, and Insights.

2. Executive Summary

Table 1 summarizes the main KPIs from the analysis.

Metric	Value
Total Revenue (Completed)	\$252,227.72
Total Orders (Completed)	3,220
Average Order Value	\$78.33
Total Revenue (All Status)	\$306,842.22
% Revenue at Risk	17.8%

Table 1. Key KPIs from the Sales Performance Summary

3. Analysis Steps (Tutorial)

3.1 Creating the First Pivot Table

All the Top Products, Top Categories, Monthly Revenue, Regional Performance, and Customer Segments analyses are built from Pivot Tables that reference the data Table in the cleaned_sales_data sheet (named Table4).

How to build it:

1. Click any cell inside the data Table (cleaned_sales_data sheet).
2. Insert > PivotTable. Excel automatically detects the Table/Range (Table4).

3. Choose New Worksheet (or Existing Worksheet if placing it in the already-created pivot_tables sheet). Click OK.

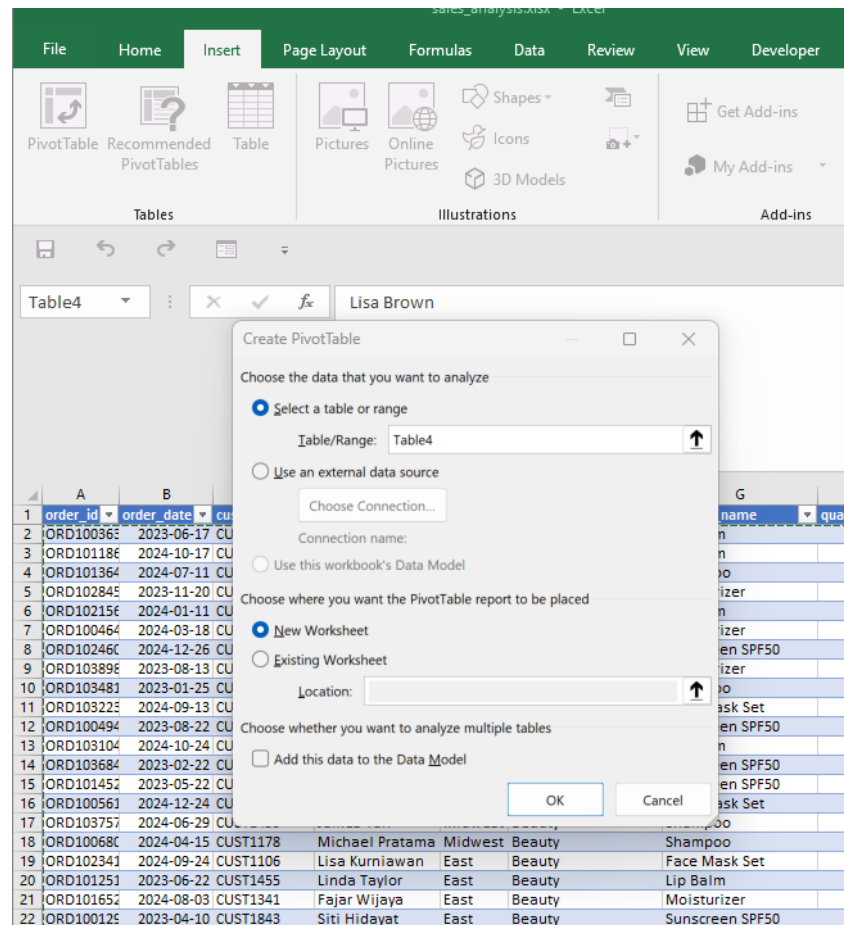


Figure 1. The Create PivotTable dialog — data source Table4, placed in a new worksheet

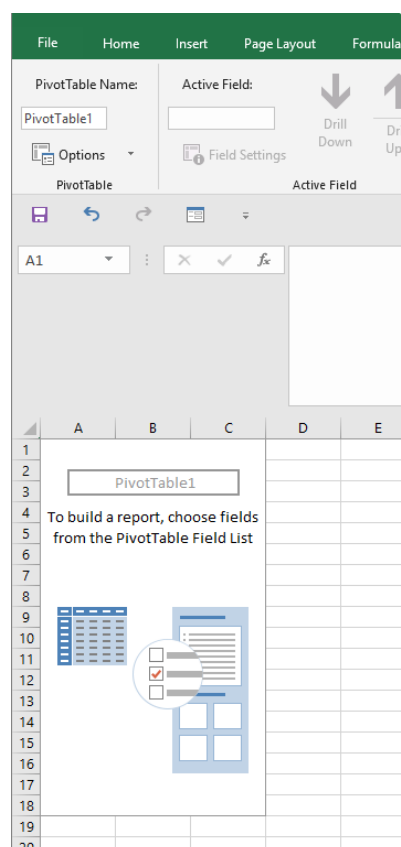


Figure 2. An empty Pivot Table before any fields are added — the field list appears in the right-hand panel

4. Drag the desired fields into the Rows and Values areas in the PivotTable Fields panel on the right. This combination of fields is what differentiates each pivot table (explained section by section below).

3.2 Top 10 Products by Revenue

1. Drag product_name into Rows, revenue into Values (automatically becomes Sum of revenue).

	A	B
1	Row Labels	Sum of revenue
2	Business Strategy Book	3326.066429
3	Cookbook	3695.323578
4	Self-Help Guide	3742.939464
5	Fiction Novel	3819.53
6	Sunscreen SPF50	4264.666863
7	Shampoo	4515.153604
8	Children's Book	4574.09405
9	Moisturizer	4916.480776
10	Face Mask Set	4987.59
11	Winter Jacket	5152.685789
12	Lip Balm	5217.873
13	Cotton T-Shirt	5616.26
14	Action Figure	5845.642621
15	RC Car	6064.454259
16	Building Blocks	6120.448077
17	Puzzle 1000pc	6185.571481
18	Wool Socks	6229.931375
19	Running Shorts	6288.79325
20	Denim Jeans	6378.182973
21	Hoodie	6480.8525
22	Hiking Backpack	6556.623636
23	Board Game	7024.31708
24	Jump Rope	7076.59
25	Baseball Cap	7138.35
26	Water Bottle	7207.49925
27	Wireless Earbuds	8143.47
28	Coffee Maker	8161.383165
29	Yoga Mat	8185.853125
30	Blender	8188.864648
31	Cutting Board	8415.375753
32	Wireless Mouse	8673.886452
33	Throw Pillow	8870.805172
34	Storage Containers	8998.935556
35	LED Desk Lamp	9069.239643
36	Bluetooth Speaker	9186.68
37	Resistance Bands	9483.796634
38	Laptop Stand	9589.976133
39	Non-Stick Pan	9605.929667
40	USB-C Cable	9894.285758
41	Camping Tent	10146.6343
42	Power Bank	10494.44957
43	Smartphone Case	11621.86466
44	Webcam	11684.86917
45	Grand Total	306842.2194
46		

Figure 3. All products with their Sum of revenue, before filtering to Top 10

2. Right-click any revenue figure → Sort → Sort Largest to Smallest.
3. Click the filter dropdown on Row Labels → Value Filters → Top 10...

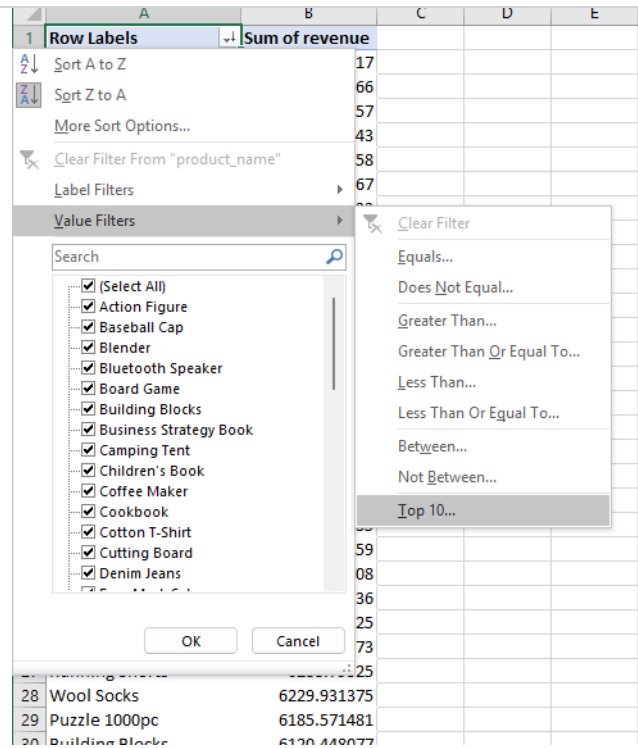


Figure 4. The Value Filters > Top 10 menu on the Row Labels dropdown

4. Make sure the filter is set to “Top 10 Items” based on Sum of revenue, then click OK.

Top 10 Products by Revenue	
Row Labels	Sum of revenue
Webcam	11684.87
Smartphone Case	11621.86
Power Bank	10494.45
Camping Tent	10146.63
USB-C Cable	9894.29
Non-Stick Pan	9605.93
Laptop Stand	9589.98
Resistance Band	9483.80
Bluetooth Speaker	9186.68
LED Desk Lamp	9069.24
Grand Total	100777.73

Figure 5. Final result: Top 10 Products by Revenue, led by Webcam (\$11,684.87)

3.3 Top Categories

The method is the same as Top Products, but the Rows field is swapped for category, and there's no need for a Top 10 filter since there are only 7 categories in total.

1. Create a new pivot table: category into Rows, revenue into Values.
2. Sort Largest to Smallest.

Top Categories	
Row Labels	Sum of revenue
Electronics	79289.48
Home & Kitchen	61310.53
Sports & Outdoors	48657.00
Clothing	43285.06
Toys	31240.43
Beauty	23901.76
Books	19157.95
Grand Total	306842.22

Figure 6. Top Categories — Electronics leads with \$79,289.48

3.4 Monthly Revenue Trend & Regional Performance

Monthly Revenue Trend

1. Create a new pivot table: order_date into Rows, revenue into Values.
2. Excel automatically offers to group the dates (a Group dialog) as soon as order_date is dragged into Rows. Check only Months and Years (uncheck Quarters if it appears), so the result is Year → Month directly, without a Quarter layer in between.

Regional Performance

1. Create a new pivot table: region into Rows. Drag revenue into Values (Sum), then drag order_id into Values a second time and change its Value Field Settings to Count — giving 2 metrics at once: total revenue and order count per region.

Top 10 Products by Revenue		Top Categories	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
Webcam	11684.87	Electronics	79289.48
Smartphone C	11621.86	Home & Kitchen	61310.53
Power Bank	10494.45	Sports & Outdoors	48657.00
Camping Tent	10146.63	Clothing	43285.06
USB-C Cable	9894.29	Toys	31240.43
Non-Stick Pan	9605.93	Beauty	23901.76
Laptop Stand	9589.98	Books	19157.95
Resistance Bar	9483.80	Grand Total	306842.22
Bluetooth Spei	9186.68		
LED Desk Lamp	9069.24		
Grand Total	100777.73		
Monthly Revenue Trend			
Row Labels	Sum of revenue		
2023	157565.78		
Jan	12576.92		
Feb	13438.69		
Mar	16241.61		
Apr	12727.19		
May	14949.47		
Jun	9447.58		
Jul	12913.82		
Aug	14167.22		
Sep	14114.10		
Oct	13900.84		
Nov	12892.05		
Dec	10196.29		
2024	149276.44		
Jan	11348.85		
Feb	11352.15		
Mar	12558.72		
Apr	10886.72		
May	14431.93		
Jun	10747.39		
Jul	12179.76		
Aug	13702.97		
Sep	10987.13		
Oct	13284.38		
Nov	13760.26		
Dec	14036.19		
Grand Total	306842.22		

Figure 7. Monthly Revenue Trend (Year → Month, no Quarter) alongside Top Products & Top Categories

	B	C	D	E	F	G
	Top 10 Products by Revenue		Top Categories			
	Row Labels	Sum of revenue	Row Labels	Sum of revenue		
	Webcam	11684.87	Electronics	79289.48		
	Smartphone C	11621.86	Home & Kitche	61310.53		
	Power Bank	10494.45	Sports & Outdc	48657.00		
	Camping Tent	10146.63	Clothing	43285.06		
	USB-C Cable	9894.29	Toys	31240.43		
	Non-Stick Pan	9605.93	Beauty	23901.76		
	Laptop Stand	9589.98	Books	19157.95		
	Resistance Bar	9483.80	Grand Total	306842.22		
	Bluetooth Spe	9186.68				
	LED Desk Lamp	9069.24				
	Grand Total	100777.73				
	Monthly Revenue Trend					
	Row Labels	Sum of revenue	Row Labels	Sum of revenue	Count of order_id	
	2023	157565.78	East	229821.11	2926	
	Jan	12576.92	Midwest	59797.74	784	
	Feb	13438.69	South	13856.29	182	
	Mar	16241.61	West	3367.08	41	
	Apr	12727.19	Grand Total	306842.22	3933	
	May	14949.47				
	Jun	9447.58				
	Jul	12913.82				
	Aug	14167.22				
	Sep	14114.10				
	Oct	13900.84				
	Nov	12892.05				
	Dec	10196.29				
	2024	149276.44				
	Jan	11348.85				
	Feb	11352.15				
	Mar	12558.72				
	Apr	10886.72				
	May	14431.93				
	Jun	10747.39				
	Jul	12179.76				
	Aug	13702.97				
	Sep	10987.13				
	Oct	13284.38				
	Nov	13760.26				
	Dec	14036.19				
	Grand Total	306842.22				

Figure 8. Regional Performance with 2 columns (Sum of revenue and Count of order_id)

An insight that jumps out right away: the East region dominates drastically — \$229,821 of \$306,842 total revenue (75%), and 2,926 of 3,933 orders (74%). The West region contributes just \$3,367 (1%).

3.5 Customer Segments

Unlike the previous 4 pivot tables, Customer Segments needs 2 helper columns in the data sheet first before its pivot table can be built — because “segment” isn’t a column available directly in the raw data.

Step 1 — Count transactions per customer (COUNTIFS)

Create a new column customer_order_count in the cleaned_sales_data sheet:

```
=COUNTIFS($C$2:$C$3934,[@[customer_id]])
```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	order_id	order_date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	
92	ORD102840	2023-06-26	CUST1138	Fajar Anderson	East	Beauty	Sunscreen SPF50	3	9.70	Completed	Debit Card	29.10	7	
128	ORD102068	2024-04-10	CUST1677	Fajar Anderson	Midwest	Sports & Outdoors	Hiking Backpack	2	10.42	Completed	Credit Card	20.84	3	
263	ORD100475	2023-03-04	CUST1097	Fajar Anderson	East	Beauty	Lip Balm	1	12.73	Returned	Gift Card	12.73	6	
716	ORD100513	2023-12-07	CUST1097	Fajar Anderson	East	Books	Self-Help Guide	1	19.51	Completed	Debit Card	19.51	6	
1071	ORD100813	2023-01-25	CUST1677	Fajar Anderson	Midwest	Beauty	Face Mask Set	1	24.50	Completed	Gift Card	24.50	3	
1211	ORD103881	2024-06-25	CUST1097	Fajar Anderson	East	Sports & Outdoors	Hiking Backpack	2	26.14	Cancelled	Debit Card	52.28	6	
1403	ORD101727	2024-07-23	CUST1097	Fajar Anderson	East	Beauty	Face Mask Set	1	28.97	Completed	Gift Card	28.97	6	
1922	ORD100951	2024-02-02	CUST1677	Fajar Anderson	Midwest	Sports & Outdoors	Yoga Mat	2	37.46	Completed	Unknown	74.92	3	
2204	ORD101568	2024-09-11	CUST1138	Fajar Anderson	East	Beauty	Moisturizer	2	42.89	Completed	PayPal	85.78	7	
2358	ORD102509	2024-11-18	CUST1138	Fajar Anderson	East	Sports & Outdoors	Resistance Bands	3	46.25	Completed	Debit Card	138.75	7	
3499	ORD103520	2023-07-10	CUST1138	Fajar Anderson	East	Sports & Outdoors	Camping Tent	1	50.21	Completed	PayPal	50.21	7	
2812	ORD101711	2023-05-19	CUST1138	Fajar Anderson	East	Toys	Building Blocks	4	59.18	Completed	Gift Card	236.72	7	
3619	ORD101572	2023-06-08	CUST1097	Fajar Anderson	East	Home & Kitchen	Non-Stick Pan	1	99.63	Pending	Gift Card	99.63	6	
3655	ORD100066	2024-09-27	CUST1138	Fajar Anderson	East	Home & Kitchen	Coffee Maker	1	102.95	Completed	Debit Card	102.95	7	
3739	ORD101265	2024-10-20	CUST1097	Fajar Anderson	East	Electronics	Wireless Earbuds	1	112.31	Cancelled	Gift Card	112.31	6	

Figure 9. The customer_order_count column — counting how many times each customer_id appears across the entire dataset

Step 2 — Classify New vs Repeat Customer (IF)

Create a new column `customer_segment`, using the result from `customer_order_count`:

```
=IF([@[customer_order_count]]=1,"New Customer","Repeat Customer")
```

B	C	D	E	F	G	H	I	J	K	L	M	N
date	customer_id	customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	customer_segment
24-06-05	CUST1081	Maria Garcia	East	Toys	Action Figure	1	10.17	Completed	Debit Card	10.17	1	New Customer
24-01-15	CUST1692	Ayu Chen	East	Books	Fiction Novel	1	10.39	Completed	Debit Card	10.39	1	New Customer
24-01-03	CUST1429	Rina Hidayat	East	Books	Children's Book	1	11.54	Cancelled	Gift Card	11.54	1	New Customer
24-01-03	CUST1204	Dewi Garcia	West	Beauty	Lip Balm	1	13.03	Returned	Credit Card	13.03	1	New Customer
23-10-10	CUST1792	Linda Sari	Midwest	Books	Business Strategy Boc	1	13.83	Completed	PayPal	13.83	1	New Customer
24-01-15	CUST1208	Sarah Pratama	East	Home & Kitchen	LED Desk Lamp	1	13.97	Completed	PayPal	13.97	1	New Customer
24-06-07	CUST1333	Andi Johnson	East	Sports & Outdoors	Camping Tent	2	16.35	Completed	Credit Card	32.70	1	New Customer
23-12-20	CUST1408	Fajar Wijaya	East	Toys	Board Game	3	16.36	Completed	Debit Card	49.08	1	New Customer
24-01-06	CUST1539	Sarah Salim	East	Books	Cookbook	1	16.91	Completed	Unknown	16.91	1	New Customer
23-08-08	CUST1358	Andi Yusuf	East	Beauty	Lip Balm	2	18.08	Completed	Debit Card	36.16	1	New Customer
24-12-03	CUST1518	Siti Salim	East	Books	Self-Help Guide	1	18.65	Completed	Gift Card	18.65	1	New Customer
24-05-25	CUST1686	Rina Garcia	West	Toys	RC Car	4	21.29	Completed	Credit Card	85.16	1	New Customer
24-08-06	CUST1331	Unknown Customer	West	Beauty	Moisturizer	1	22.77	Cancelled	Debit Card	22.77	1	New Customer
24-07-14	CUST1782	Budi Brown	Midwest	Books	Self-Help Guide	1	22.97	Completed	Debit Card	22.97	1	New Customer
23-02-24	CUST1044	David Smith	West	Beauty	Face Mask Set	1	23.36	Cancelled	PayPal	23.36	1	New Customer
24-06-22	CUST1371	Sarah Taylor	Midwest	Books	Cookbook	1	23.67	Pending	PayPal	23.67	1	New Customer
23-02-31	CUST1557	Rina Lee	Midwest	Books	Fiction Novel	2	24.08	Completed	PayPal	48.16	1	New Customer
23-06-21	CUST1805	John Chen	South	Home & Kitchen	Storage Containers	2	25.03	Completed	Gift Card	50.06	1	New Customer
24-11-02	CUST1749	Unknown Customer	West	Toys	Board Game	1	25.57	Completed	Gift Card	25.57	1	New Customer
24-09-02	CUST1864	John Anderson	South	Books	Business Strategy Boc	1	27.47	Completed	Credit Card	27.47	1	New Customer
23-03-23	CUST1698	James Johnson	South	Beauty	Face Mask Set	1	27.58	Completed	PayPal	27.58	1	New Customer
23-12-16	CUST1598	Nina Halim	Midwest	Beauty	Shampoo	1	28.02	Completed	Credit Card	28.02	1	New Customer
24-05-22	CUST1706	Putri Hidayat	South	Books	Children's Book	1	28.76	Completed	Credit Card	28.76	1	New Customer
23-09-09	CUST1325	Ayu Kurniawan	East	Clothing	Baseball Cap	1	28.87	Completed	Gift Card	28.87	1	New Customer
24-04-21	CUST1556	Budi Wilson	East	Books	Fiction Novel	1	30.45	Completed	Debit Card	30.45	1	New Customer
24-05-20	CUST1570	Siti Wijaya	East	Beauty	Moisturizer	2	32.73	Completed	Debit Card	65.46	1	New Customer
23-01-13	CUST1767	Andi Smith	West	Toys	Building Blocks	1	34.87	Completed	Debit Card	34.87	1	New Customer
23-12-05	CUST1228	Dewi Santos	Midwest	Sports & Outdoors	Resistance Bands	1	35.69	Completed	PayPal	35.69	1	New Customer
23-05-27	CUST1530	Nina Garcia	East	Beauty	Moisturizer	1	35.83	Completed	Gift Card	35.83	1	New Customer
24-12-19	CUST1489	Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	108.87	1	New Customer
23-05-04	CUST1326	Sarah Kurniawan	South	Beauty	Lip Balm	1	37.09	Completed	Gift Card	37.09	1	New Customer
24-12-16	CUST1538	Dewi Kurniawan	East	Toys	Action Figure	1	40.42	Completed	Credit Card	40.42	1	New Customer
23-03-19	CUST1727	Nina Brown	Midwest	Sports & Outdoors	Resistance Bands	2	52.35	Completed	Debit Card	104.70	1	New Customer
23-11-12	CUST1083	Ahmad Taylor	East	Clothing	Winter Jacket	1	53.26	Completed	Credit Card	53.26	1	New Customer
23-11-22	CUST1536	Maria Hidayat	West	Clothing	Running Shorts	2	58.04	Completed	Credit Card	116.08	1	New Customer
24-09-24	CUST1797	Ahmad Brown	South	Toys	RC Car	3	58.06	Completed	Debit Card	174.18	1	New Customer

Figure 10. The `customer_segment` formula applied — customers with 1 transaction are marked New Customer, the rest Repeat Customer

D	E	F	G	H	I	J	K	L	M	N
customer_name	region	category	product_name	quantity	unit_price	order_status	payment_method	revenue	customer_order_count	customer_segment
Maria Garcia	East	Toys	Action Figure	1	10.17	Completed	Debit Card	10.17	1	New Customer
Ayu Chen	East	Books	Fiction Novel	1	10.39	Completed	Debit Card	10.39	1	New Customer
Rina Hidayat	East	Books	Children's Book	1	11.54	Cancelled	Gift Card	11.54	1	New Customer
Dewi Garcia	West	Beauty	Lip Balm	1	13.03	Returned	Credit Card	13.03	1	New Customer
Linda Sari	Midwest	Books	Business Strategy Boc	1	13.83	Completed	PayPal	13.83	1	New Customer
Sarah Pratama	East	Home & Kitchen	LED Desk Lamp	1	13.97	Completed	PayPal	13.97	1	New Customer
Andi Johnson	East	Sports & Outdoors	Camping Tent	2	16.35	Completed	Credit Card	32.70	1	New Customer
Fajar Wijaya	East	Toys	Board Game	3	16.36	Completed	Debit Card	49.08	1	New Customer
Sarah Salim	East	Books	Cookbook	1	16.91	Completed	Unknown	16.91	1	New Customer
Andi Yusuf	East	Beauty	Lip Balm	2	18.08	Completed	Debit Card	36.16	1	New Customer
Siti Salim	East	Books	Self-Help Guide	1	18.65	Completed	Gift Card	18.65	1	New Customer
Rina Garcia	West	Toys	RC Car	4	21.29	Completed	Credit Card	85.16	1	New Customer
Unknown Customer	West	Beauty	Moisturizer	1	22.77	Cancelled	Debit Card	22.77	1	New Customer
Budi Brown	Midwest	Books	Self-Help Guide	1	22.97	Completed	Debit Card	22.97	1	New Customer
David Smith	West	Beauty	Face Mask Set	1	23.36	Cancelled	PayPal	23.36	1	New Customer
Sarah Taylor	Midwest	Books	Cookbook	1	23.67	Pending	PayPal	23.67	1	New Customer
Rina Lee	Midwest	Books	Fiction Novel	2	24.08	Completed	PayPal	48.16	1	New Customer
John Chen	South	Home & Kitchen	Storage Containers	2	25.03	Completed	Gift Card	50.06	1	New Customer
Unknown Customer	West	Toys	Board Game	1	25.57	Completed	Gift Card	25.57	1	New Customer
John Anderson	South	Books	Business Strategy Boc	1	27.47	Completed	Credit Card	27.47	1	New Customer
James Johnson	South	Beauty	Face Mask Set	1	27.58	Completed	PayPal	27.58	1	New Customer
Nina Halim	Midwest	Beauty	Shampoo	1	28.02	Completed	Credit Card	28.02	1	New Customer
Putri Hidayat	South	Books	Children's Book	1	28.76	Completed	Credit Card	28.76	1	New Customer
Ayu Kurniawan	East	Clothing	Baseball Cap	1	28.87	Completed	Gift Card	28.87	1	New Customer
Budi Wilson	East	Books	Fiction Novel	1	30.45	Completed	Debit Card	30.45	1	New Customer
Siti Wijaya	East	Beauty	Moisturizer	2	32.73	Completed	Debit Card	65.46	1	New Customer
Andi Smith	West	Toys	Building Blocks	1	34.87	Completed	Debit Card	34.87	1	New Customer
Dewi Santos	Midwest	Sports & Outdoors	Resistance Bands	1	35.69	Completed	PayPal	35.69	1	New Customer
Nina Garcia	East	Beauty	Moisturizer	1	35.83	Completed	Gift Card	35.83	1	New Customer
Nina Salim	Midwest	Clothing	Running Shorts	3	36.29	Completed	Gift Card	108.87	1	New Customer
Sarah Kurniawan	South	Beauty	Lip Balm	1	37.09	Completed	Gift Card	37.09	1	New Customer
Dewi Kurniawan	East	Toys	Action Figure	1	40.42	Completed	Credit Card	40.42	1	New Customer
Nina Brown	Midwest	Sports & Outdoors	Resistance Bands	2	52.35	Completed	Debit Card	104.70	1	New Customer
Ahmad Taylor	East	Clothing	Winter Jacket	1	53.26	Completed	Credit Card	53.26	1	New Customer
Maria Hidayat	West	Clothing	Running Shorts	2	58.04	Completed	Credit Card	116.08	1	New Customer

Figure 12. The completed `customer_segment` column, filtered to show New Customer only as verification

Step 3 — Build the Customer Segments Pivot Table

1. Create a new pivot table: `customer_segment` into Rows, revenue into Values (Sum), and `order_id` into Values again (Count).

Customer Segments		
Row Labels	Sum of revenue	Count of Orders
New Customer	3427.88	48
Repeat Customer	303414.34	3885
Grand Total	306842.22	3933

Figure 11. Customer Segments result — Repeat Customers dominate at 3,885 of 3,933 orders (98.8%)

3.6 Overview: All Pivot Tables

All five pivot tables (Top Products, Top Categories, Monthly Revenue Trend, Regional Performance, Customer Segments) are arranged side by side in the pivot_labels sheet so they read as one compact mini-dashboard.

Top 10 Products by Revenue		Top Categories	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
Webcam	11684.87	Electronics	79289.48
Smartphone Case	11621.86	Home & Kitchen	61310.53
Power Bank	10494.45	Sports & Outdoors	48657.00
Camping Tent	10146.63	Clothing	43285.06
USB-C Cable	9894.29	Toys	31240.43
Non-Stick Pan	9605.93	Beauty	23901.76
Laptop Stand	9589.98	Books	19157.95
Resistance Bands	9483.80	Grand Total	306842.22
Bluetooth Speaker	9186.68		
LED Desk Lamp	9069.24		
Grand Total	100777.73		
Monthly Revenue Trend		Regional Performance	
Row Labels	Sum of revenue	Row Labels	Sum of revenue
2023	157565.78	East	229821.11
Jan	12576.92	Midwest	59797.74
Feb	13438.69	South	13856.29
Mar	16241.61	West	3367.08
Apr	12727.19	Grand Total	306842.22
May	14949.47		
Jun	9447.58		
Jul	12913.82		
Aug	14167.22		
Sep	14114.10		
Oct	13900.84		
Nov	12892.05		
Dec	10196.29		
2024	149276.44		
Jan	11348.85		
Feb	11352.15		
Mar	12558.72		
Apr	10886.72		
May	14431.93		
Jun	10747.39		
Jul	12179.76		
Aug	13702.97		
Sep	10987.13		
Oct	13284.38		
Nov	13760.26		
Dec	14036.19		
Grand Total	306842.22		
Customer Segments			
Row Labels	Sum of revenue	Count of Orders	
New Customer	3427.88	48	
Repeat Customer	303414.34	3885	
Grand Total	306842.22	3933	

Figure 13. The final layout of all five pivot tables in the pivot_labels sheet

3.7 Sales Performance Summary (SUMIFS, COUNTIF)

The main KPIs are calculated in the Summary sheet rather than simply pulled from a pivot table's Grand Total — so they can be filtered specifically to Completed orders (revenue that has actually been realized, excluding Cancelled/Returned/Pending).

Total Revenue (Completed):
`=SUMIFS(Table4[revenue],Table4[order_status],"Completed")`

Total Orders (Completed):
`=COUNTIF(Table4[order_status],"Completed")`

Average Order Value:
`=D3/D4`

Total Revenue (All Status):
`=SUM(Table4[revenue])`

Figure 14. Sales Performance Summary — Total Revenue Completed \$252,227.72, AOV \$78.33, Revenue at Risk 17.8%

3.8 Customer Lookup Tool (XLOOKUP vs INDEX/MATCH)

If your Excel supports XLOOKUP (Microsoft 365 / Excel 2021+)

1. Type =XLOOKUP(then click cell D12 (the lookup value, i.e. the customer_id typed by the user).
2. Comma, then click-drag/select the entire customer_id column in the Table (Table4[customer_id]) — this is the "lookup_array", where XLOOKUP searches for a match.
3. Comma, then click-drag/select the column whose value you want returned, e.g. customer_name (Table4[customer_name]) — this is the "return_array".
4. Close the parenthesis, Enter.

```
Region:
=XLLOOKUP(D12,Table4[customer_id],Table4[region])

Segment:
=XLLOOKUP(D12,Table4[customer_id],Table4[customer_segment])

Total Orders:
=XLLOOKUP(D12,Table4[customer_id],Table4[customer_order count])
```

The version actually used in this file: INDEX + MATCH

Since XLOOKUP wasn't available, INDEX + MATCH was used instead, giving an identical result. The formula pattern is the same for all four fields shown by the Customer Lookup Tool — just swap the column being indexed:

```
Customer Name:
=INDEX(Table4[customer_name],MATCH($D$12,Table4[customer_id],0))

Region:
=INDEX(Table4[region],MATCH($D$12,Table4[customer_id],0))

Segment:
=INDEX(Table4[customer_segment],MATCH($D$12,Table4[customer_id],0))

Total Orders:
=INDEX(Table4[customer_order_count],MATCH($D$12,Table4[customer_id],0))
```

How to read the formula: MATCH finds the ROW POSITION where customer_id equals D12, then INDEX pulls the value from the chosen column (customer_name / region / customer_segment / customer_order_count) at that row position. The 0 at the end of MATCH is mandatory — it means “find an EXACT match”, not an approximate one. Without it, the lookup result could be wrong since customer_id isn't sorted.

Sales Performance Summary	
Total Revenue (Completed)	\$252,227.72
Total Orders (Completed)	3220
Average Order Value	\$78.33
Total Revenue (All Status)	\$306,842.22
% Revenue at Risk	17.80%
Customer Lookup Tool	
Enter Customer ID	CUST1138
Customer Name	Fajar Anderson

Figure 15. Customer Lookup Tool — enter a customer_id (CUST1138), the customer's name is displayed automatically (Fajar Anderson)

3.9 Charts (PivotChart)

Four charts were built directly from the pivot tables using PivotChart (not a regular Chart), so they update automatically whenever the underlying pivot table data changes.

1. Click inside the source pivot table → Insert → PivotChart → choose a chart type (Line for time trends, Bar for rankings, Pie for proportions).
2. For the Regional Performance chart, which combines 2 metrics on different scales (revenue and order count), a Combo Chart was used: revenue as a Column on the primary axis, Count of order_id as a Line on the Secondary Axis — so both are proportionally visible, with neither hidden by the other's scale.
3. Field Buttons (the filter dropdowns that automatically attach to a chart's corner) were hidden via PivotChart Analyze > Field Buttons > Hide All, for a cleaner look in the report.

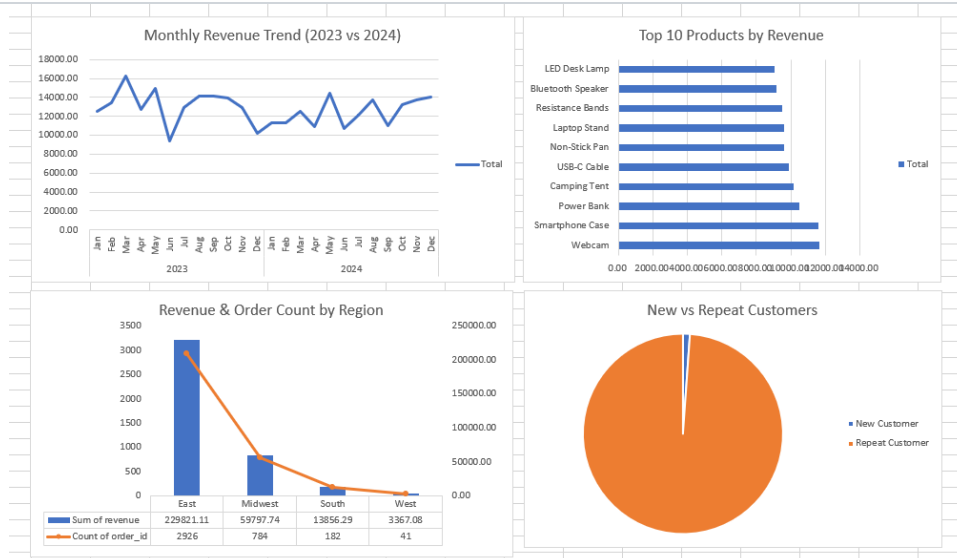


Figure 17. 4-chart dashboard: Monthly Revenue Trend, Top 10 Products, Revenue & Order Count by Region, New vs Repeat Customers

3.10 Progress Checklist

12	Day 2	Excel Analysis	Hitung Total Revenue	Excel	SELESAI
13	Day 2	Excel Analysis	Hitung Total Orders	Excel	SELESAI
14	Day 2	Excel Analysis	Hitung Average Order Value	Excel	SELESAI
15	Day 2	Excel Analysis	Cari Top Products	Excel	SELESAI
16	Day 2	Excel Analysis	Cari Top Categories	Excel	SELESAI
17	Day 2	Excel Analysis	Buat Monthly Revenue	Excel	SELESAI
18	Day 2	Excel Analysis	Buat Regional Performance	Excel	SELESAI
19	Day 2	Excel Analysis	Buat Customer Segments	Excel	SELESAI
20	Day 2	Excel Analysis	Pakai Pivot Table	Excel	SELESAI
21	Day 2	Excel Analysis	Pakai XLOOKUP/INDEX MATCH	Excel	SELESAI
22	Day 2	Excel Analysis	Pakai SUMIFS	Excel	SELESAI
23	Day 2	Excel Analysis	Pakai COUNTIFS	Excel	SELESAI
24	Day 2	Excel Analysis	Pakai IF	Excel	SELESAI
25	Day 2	Excel Analysis	Buat charts	Excel	
26	Day 2	Excel Analysis	Tulis 5-10 business insights	Excel	
27	Day 2	Excel Analysis	Simpan sebagai Sales_Analysis.xlsx	Excel	

Figure 16. Day 2 checklist at the stage where the basic formulas (SUMIFS, COUNTIFS, IF) are done, with Charts and Insights still to follow

4. Business Insights

The eight insights below were drawn from all the figures and patterns found throughout the analysis above.

1. Extreme Dependence on the East Region

The East region accounts for 75% of total revenue (\$229,821 of \$306,842) and 74% of total orders (2,926 of 3,933), while West contributes just 1% (\$3,367, 41 orders). This extreme imbalance shows the business is heavily reliant on a single region — a major risk if East's performance declines.

2. 17.8% of Revenue Is at Risk of Not Being Realized

Of the \$306,842 gross revenue, only \$252,228 (82.2%) comes from orders with a Completed status. The remaining 17.8% (\$54,614) comes from Cancelled, Returned, or Pending orders — revenue that isn't guaranteed to actually convert into cash.

3. Customer Base Dominated by Repeat Customers, Weak New Acquisition

98.8% of orders (3,885 of 3,933) come from Repeat Customers, with only 48 orders from New Customers. This signals strong customer retention, but on the flip side points to very weak new-customer acquisition that needs to be addressed.

4. Electronics Is the Dominant Category and Product Driver

Electronics is the highest-revenue category (\$79,289, 26% of the total), followed by Home & Kitchen (\$61,311). Three of the top five best-selling products (Webcam, Smartphone Case, Power Bank) also come from Electronics — this category is the main sales driver.

5. Books Is the Weakest-Performing Category

Books has the lowest revenue of any category (\$19,158), less than a quarter of Electronics' revenue. This category could be worth re-evaluating in terms of product range or promotional strategy.

6. No Consistent Seasonal Pattern

Monthly revenue fluctuates with no clear seasonal pattern — peaking in March 2023 (\$16,242), then dropping to its lowest point in June 2023 (\$9,448). This volatility suggests revenue is driven more by random or operational factors than by seasonality.

7. Annual Revenue Slightly Declined, Not Grew

Total revenue dipped slightly from \$157,566 (2023) to \$149,276 (2024), a year-over-year decrease of about 5.3%. Small as it is, this is still a negative trend worth monitoring, not growth.

8. AOV Is Relatively Low, Suggesting a Bundling/Cross-Sell Opportunity

Average Order Value is just \$78.33 across 3,220 Completed orders, consistent with a quantity pattern where most orders are just 1 unit. There's an opportunity to raise AOV through bundling or cross-selling complementary products.

5. Day 2 Checklist Summary

Task	Status	Notes
Calculate Total Revenue	Done	SUMIFS, filtered by order_status = Completed
Calculate Total Orders	Done	COUNTIF, filtered by order_status = Completed
Calculate Average Order Value	Done	Total Revenue ÷ Total Orders
Find Top Products	Done	Pivot Table + Top 10 filter
Find Top Categories	Done	Pivot Table, 7 categories
Build Monthly Revenue	Done	Pivot Table, grouped by Year + Month
Build Regional Performance	Done	Pivot Table + Secondary Axis chart
Build Customer Segments	Done	Helper columns via COUNTIFS + IF, then a Pivot Table
Use Pivot Table	Done	5 pivot tables used in total
Use XLOOKUP / INDEX-MATCH	Done	Substituted with INDEX/MATCH — the Excel version did not support XLOOKUP
Use SUMIFS	Done	Total Revenue (Completed)
Use COUNTIFS	Done	customer_order_count
Use IF	Done	customer_segment
Build charts	Done	4 PivotCharts in the Charts sheet
Write 5-10 business insights	Done	8 insights in the Insights sheet
Save as Sales_Analysis.xlsx	Done	3,933 clean data rows

Table 2. Status of every Day 2 Excel Analysis task

6. Conclusion

The Day 2 analysis produced Sales_Analysis.xlsx, containing 5 pivot tables, 4 interactive charts, a KPI summary built with SUMIFS/COUNTIF, a Customer Lookup Tool built with INDEX/MATCH, and 8 business insights backed by concrete figures. The entire process was built on top of the clean 3,933-row dataset from Day 1, so the figures reported here stay consistent with the earlier stages.

The most significant findings: extreme dependence on the East region (75% of revenue), a potential 17.8% revenue loss from non-Completed orders, and very weak new-customer acquisition (only 1.2% of total orders). These three points are strong candidates for deeper discussion in Day 6 (Business Case Study).

This Sales_Analysis.xlsx file becomes the input for Day 3 (SQL queries), Day 4 (Python EDA), and Day 5 (Tableau dashboard) — in particular, the KPI figures and insights above will be used as a cross-check/validation reference at those stages.